

Project ID: 40

Project Title

Doe-Assist (Intelligent Parameter Reduction For Experimental Design)

Client Name

Rishi Ravindra Naik

Group Capacity

3 groups

Project Tags

Artificial Intelligence (Machine/Deep Learning, NLP), Software Development, Big data Analytics and Visualization

Company/Organization Name (Including Department)

School of Chemical Engineering

Project Background

Design of Experiments (DoE) is a foundational methodology in scientific research and engineering, enabling systematic investigation of the effects of multiple variables on outcomes. However, the initial phase of selecting starting parameters is often time-consuming, resource-intensive, and heavily reliant on expert intuition or trial-and-error. This inefficiency is particularly pronounced in complex domains such as solubility studies, where the parameter space can be vast and interdependent. With the exponential growth of scientific literature and experimental data available online, there is an untapped opportunity to leverage this knowledge to inform and optimize experimental design. By mining existing literature and integrating manual data entry, we can build a dynamic, self-improving system that learns from past experiments to guide future ones. The goal of this project is to develop an intelligent system that:

1. Automatically mines experimental parameters and outcomes from scientific literature and databases.
2. Allows manual input of experimental data to enrich and validate the dataset.
3. Constructs a structured database of experiments categorized by type (e.g., solubility, crystallization, etc.).
4. Trains a machine learning model that can predict optimal starting parameters for a given experimental setup.
5. Provides researchers with parameter recommendations, reducing the need for exhaustive trial-and-error and accelerating the path to successful outcomes.
6. Ultimately, the software aims to become a decision-support tool that enhances experimental efficiency, reduces material waste, and democratizes access to expert-level experimental planning.

Project Scope

The scope of this project will encompass the development of an intelligent, modular system that assists researchers in optimizing the design of experiments by reducing the number of initial parameters required. This includes building a pipeline for automated literature mining to extract experimental parameters and outcomes from scientific publications, as well as a manual data entry interface to capture unpublished or user-specific data. A structured database will be developed to store and organize this information, serving as the foundation for training machine learning models capable of predicting optimal starting parameters for specific experiment types, initially focusing on solubility studies. The system will also include a feedback mechanism to incorporate real-world experimental results, enabling continuous learning and refinement of predictions. Out of scope for the initial phase are real-time lab integration, full automation of experimental design, and deployment features.

Project Requirements

The system will begin with a Literature Mining Engine, which is responsible for automatically extracting experimental parameters and outcomes from scientific publications and online databases. This component will utilize APIs such as Semantic Scholar and CrossRef, as well as web scraping tools where necessary. Natural Language Processing (NLP) techniques will be employed to identify and extract relevant information such as experiment types, variables, and results. The engine will also capture metadata like publication source, date, and confidence scores, and will include filters to focus on specific domains such as solubility studies. To complement automated data collection, Paramine will include a Manual Data Entry Interface that allows researchers to input their own experimental data. This interface will be web-based and designed with user-friendly forms that include validation checks to ensure data consistency. It will support structured uploads (e.g., CSV or Excel files) and allow tagging of experiments by type, material, and method. User authentication and version control will be implemented to maintain data integrity and traceability. All collected data both mined and manually entered will be stored in a Centralized Experiment Database (needs to be stored locally in the users computer). This database will be designed using either a relational model (e.g., PostgreSQL) or a NoSQL approach (e.g., MongoDB), depending on the complexity and variability of the data. The schema will include fields for experiment type, parameters, outcomes, source, confidence scores, and tags. The database will be fully queryable via internal tools and APIs, and will be built with scalability, security, and regular backup mechanisms in mind. At the core of the system will be the Machine Learning Prediction Engine, which will analyze the database to predict optimal starting parameters for new experiments. This engine will use supervised or semi-supervised learning models such as Random Forests, XGBoost, or Neural Networks. Users will input the type of experiment and any known constraints, and the engine will return a ranked list of recommended parameter sets along with confidence scores. The model will be retrained periodically as new data is added, ensuring continuous improvement in prediction accuracy. To make these predictions accessible, Paramine will feature a User Query Interface. This interface will allow researchers to input their

experimental context and receive parameter suggestions. It will be available as a simple web interface or command-line tool, and will provide outputs that include not only the recommended parameters but also the rationale behind them such as references to similar past experiments. Users will also have the option to export results in formats like PDF or CSV for documentation or reporting purposes. A key feature of the system will be its Feedback and Learning Loop, which enables users to submit the outcomes of experiments conducted using Paramine's suggestions. These results will be used to update the confidence scores of existing data and to retrain the machine learning models, thereby improving the system's predictive capabilities over time. This feedback mechanism ensures that the system evolves with real-world use and becomes increasingly accurate and reliable. Finally, the entire system will be built with a Modular and Scalable Architecture. It will follow a microservices design, with each component (e.g., mining, database, ML engine) operating independently but communicating via RESTful APIs. The system will be containerized using Docker for easy deployment and maintenance, and will include logging and monitoring tools such as Prometheus and Grafana to ensure reliability and performance.

Required Skills

To successfully undertake the Paramine project, a student should possess a strong foundation in programming particularly in Python for data processing, machine learning, and web scraping as well as basic web development skills for building user interfaces and APIs. They should be familiar with machine learning concepts and tools, including model training, evaluation, and deployment, and have experience with natural language processing (NLP) techniques for extracting structured data from scientific literature. Knowledge of database systems (SQL or NoSQL) is essential for designing and managing the experimental data repository. Additionally, the student should understand the principles of experimental design, especially in scientific domains like solubility studies, to contextualize the data and predictions. Soft skills such as critical thinking, problem-solving, and effective communication are also important for collaborating with researchers, documenting progress, and iterating on feedback.

Expected Outcomes

The expected outcomes and deliverables of the project should include a fully functional software system capable of mining experimental parameters from literature, accepting manual data input, and predicting optimal starting parameters for new experiments. Deliverables will consist of the complete source code for all system components (including the literature mining engine, database, machine learning models, and user interfaces), a well-structured database schema, and a trained machine learning model with documented performance metrics. Additionally, the project will include comprehensive technical documentation detailing system architecture, setup instructions, and code annotations, as well as a user guide that explains how to input data, query the system, and interpret predictions. Optional deliverables may include a demo video, presentation slides, and a deployment-ready Docker container for easy installation and testing.

Disciplines

Software Development;Artificial Intelligence (Machine/Deep Learning, NLP);Big data Analytics and Visualization;

Other Resources

None